# Samuel Nellessen AI Safety Research Engineer

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AI safety researcher building automated red-teaming, adversarial evals, and RL-based jailbreak discovery systems for agentic LLMs.

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## Experience

**Incoming Fellow** | LASR Labs — Jul 2026 - Sep 2026

- Selected for technical AI safety fellowship in London. Expected project focus: automated red-teaming, evals, and agentic LLM safety.

**Independent Researcher & Foresight Grantee** | Foresight Institute — Mar 2026 - Present

- Awarded competitive $22.5k AI Safety Grant to direct independent research on automated red-teaming and agentic LLM safety.
- Developing an ICLR submission on split behavior in tool-using LLM agents, where models produce refusal-like text while still executing prohibited tool actions.

**Student Researcher** | KachmanLab @ Radboud University — Sep 2025 - Present

- Developing novel jailbreaking techniques and adversarial attacks on LLMs.
- Orchestrating multi-GPU distributed training (Slurm/HPC) for Clipped Importance Sampling Policy Optimization (CISPO) to test alignment boundaries using automated jailbreakers.

**ARENA 5.0 Fellow** | Alignment Research Engineer Accelerator — Apr 2025 - Jun 2025

- Selected for competitive fellowship with 4% acceptance rate, training alongside PhDs and industry professionals.
- Implemented Transformers from scratch and conducted deep dives into RLHF and interpretability.

**Research Assistant & Foresight Fellow** | Donders Institute / Foresight Institute — Feb 2024 - Jul 2025

- Architected a scalable model-fitting pipeline on Slurm HPC clusters to quantify controllability, translating theoretical frameworks into executable code.
- Selected as **Foresight Fellow** to lead computational modeling of agency, independently developing algorithms to extract latent parameters from behavioral data.

**AI Safety Grantee & Scholar** | Long-Term Future Fund / AIMM — Jul 2022 - Sep 2023

- Awarded competitive grant (<5% acceptance) and mentorship to execute independent research under Jan Hendrik Kirchner (OpenAI/Anthropic).

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## Selected Research

**Split Behavior in Tool-Using LLM Agents** — 2026

*ICLR submission in preparation. Studying cases where models produce refusal-like text while still executing prohibited or harmful tool actions. Building behavioral and mechanistic evidence that text-level refusal and action-level safety can decouple in agentic LLM systems.*

**David vs. Goliath: Verifiable Agent-to-Agent Jailbreaking via Reinforcement Learning** — 2026

*NeurIPS 2026 submission / arXiv preprint. Led a project formalizing Tag-Along Attacks, where adversaries exploit tool necessity in agentic settings. Built an RL framework for automated jailbreak discovery, with attacks transferring zero-shot to closed models such as Gemini.*

**Reasoning Capabilities of LLMs** — 2025

*EACL 2026 accepted paper from AI Safety Camp 2025. Contributed mechanistic interpretability analyses in an international research team studying internal representations of reasoning-related capabilities. Paper available on ACL Anthology.*

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## Education

**Radboud University** — Nijmegen, Netherlands

*B.Sc. in Artificial Intelligence (Honours Programme)* — 2023 - 2026 *(expected)*

- Current GPA: 4.00/4.00 (Dutch system: 9/10).

**Otto von Guericke University / Stendal University** — Germany

*B.A. Philosophy, Neuroscience & B.Sc. Psychology (Foundational Studies)* — 2020 - 2023

- Consistently ranked in top percentile (GPA $\approx$ 3.9/4.0). Transferred to Radboud to specialize in technical AI.

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## Projects

**Styx Interchange: Localizing Refusal Behavior in LLMs** 2026

*Weekend mechanistic interpretability sprint isolating refusal mechanisms in Qwen3-0.6B. Compared activation patching against input gradients using paired prompts, finding that causal refusal behavior is concentrated in specific model layers.*

**Internal Representations in SONAR Autoencoders** 2025

*ARENA 5.0 capstone. Analyzed internal representations in a text autoencoder to identify features correlated with code validity, grammaticality, and correctness-like structure.*

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## Open Source

**UK AISI `inspect_ai`:** Merged PR fixing vLLM startup behaviour.

**Prime Intellect Environments:** Merged PR for text retrieval environment. Implementing AI Safety benchmarks, e.g. AgentDojo environment.

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## Skills

**Research:** Automated red-teaming, adversarial evals, LLM jailbreaking, RL-based attack discovery, mechanistic interpretability.

**Engineering:** PyTorch, Transformers, verifiers, Slurm/HPC, multi-GPU training, AgentDojo, inspect_ai.

**Other:** Python, Pandas, MATLAB, Stan, JavaScript/React.

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## Leadership & Awards

**Technical Advisor**, Foresight AI Safety Grant Program 2024

**User Experience Lead**, Fridays for Future International 2019-2020